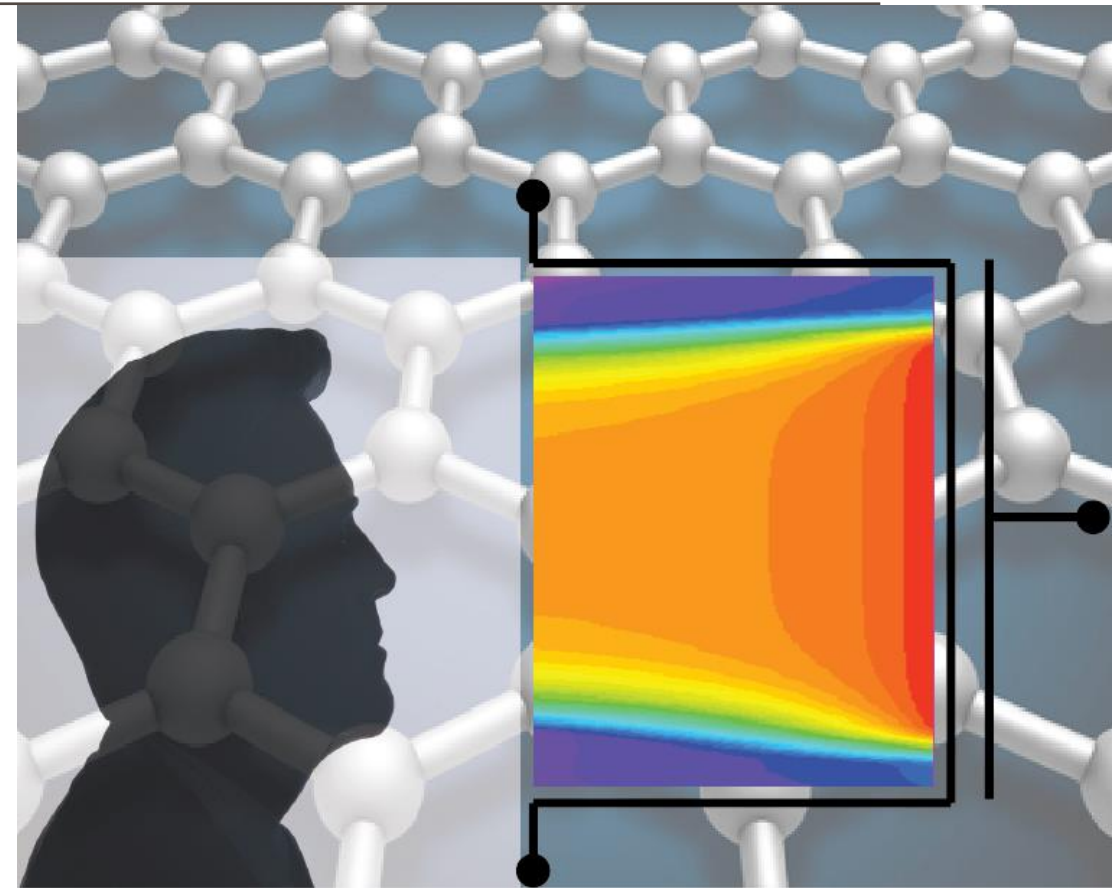


ADVANCED SOLID-STATE DEVICES

Week-3, Lecture-1

Sneh Saurabh
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Extrinsic Semiconductor in Equilibrium

N-type semiconductor

- Due to donor and acceptor atoms, distribution of electrons and holes changes
- Fermi energy level changes
- When $E_F > E_{Fi}$, electron concentration becomes greater than hole concentration (n-type) material
- Concentration of electrons: $n_0 = N_C e^{-(E_C - E_F)/kT}$

We can write electron concentration in another form:

$$\begin{aligned}
 n_0 &= N_C e^{-(E_C - E_F)/kT} \\
 &= N_C e^{-(E_C - E_{Fi} + E_{Fi} - E_F)/kT} \\
 &= N_C e^{-(E_C - E_{Fi})/kT} \times e^{(E_F - E_{Fi})/kT} = n_i e^{(E_F - E_{Fi})/kT}
 \end{aligned}$$

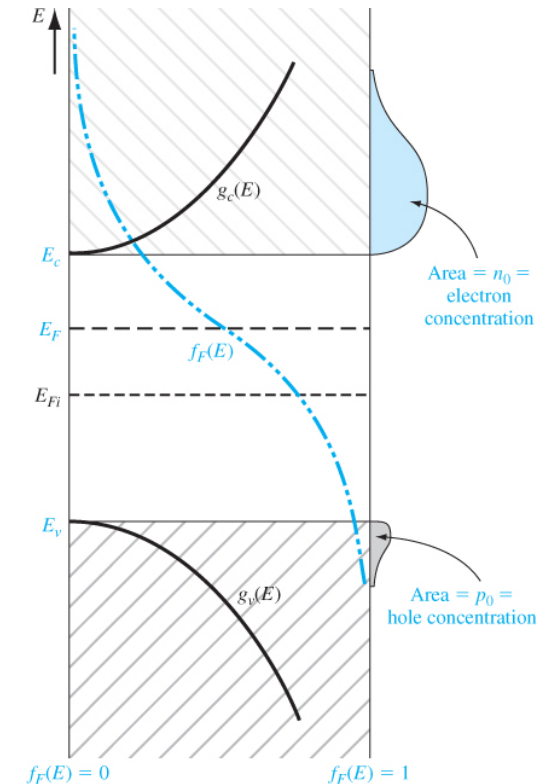


Figure 4.8 | Density of states functions, Fermi-Dirac probability function, and areas representing electron and hole concentrations for the case when E_F is above the intrinsic Fermi energy.

P-type semiconductor

- When $E_F < E_{Fi}$, hole concentration becomes greater than electron concentration (p-type) material
- Fermi energy level changes

- Concentration of holes: $p_0 = N_V e^{-(E_F - E_V)/kT}$

We can write hole concentration in another form:

$$\begin{aligned}
 p_0 &= N_V e^{-(E_F - E_V)/kT} \\
 &= N_V e^{-(E_F - E_{Fi} + E_{Fi} - E_V)/kT} \\
 &= N_V e^{-(E_{Fi} - E_V)/kT} \times e^{(E_{Fi} - E_F)/kT} = n_i e^{(E_{Fi} - E_F)/kT}
 \end{aligned}$$

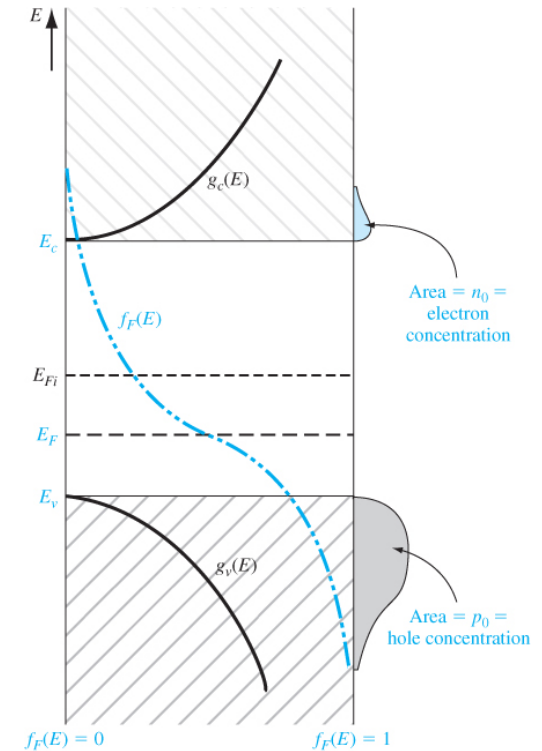


Figure 4.9 | Density of states functions, Fermi-Dirac probability function, and areas representing electron and hole concentrations for the case when E_F is below the intrinsic Fermi energy.

The product of electron and hole concentration

- Electron Concentration: $n_0 = N_C e^{-(E_C - E_F)/kT}$

- Hole concentration: $p_0 = N_V e^{-(E_F - E_V)/kT}$

$$\begin{aligned} n_0 p_0 &= N_C e^{-(E_C - E_F)/kT} \times N_V e^{-(E_F - E_V)/kT} \\ &= N_C N_V e^{-E_G/kT} = n_i^2 \end{aligned}$$

$$n_0 p_0 = n_i^2$$

- We can use this expression for computing minority carrier concentration (hole in n-type and electron in p-type)
- Be careful about the assumptions used in the derivation

Equilibrium Electron-Hole Concentration Distribution

When we add donor atoms: redistribution of charges take place

- Electron concentration increases in the conduction band
- Some electrons of the donor atoms also fall back to the valence band
- The intrinsic holes (generated thermally as electron-hole pairs) get annihilated in the valence band
- Hole concentration decreases below the intrinsic level as we add donor atoms

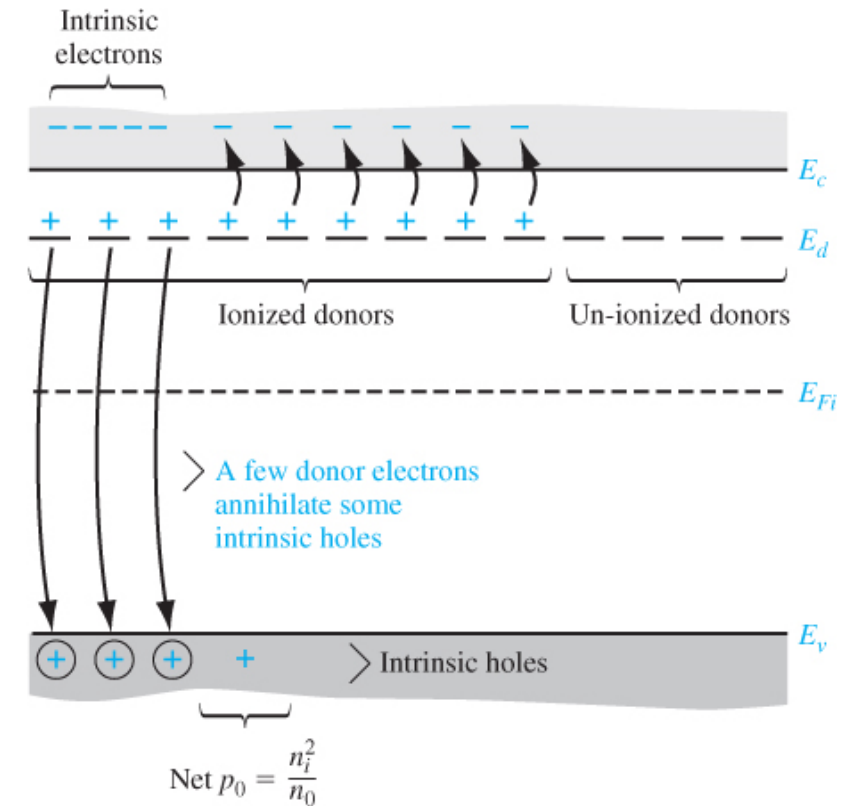


Figure 4.15 | Energy-band diagram showing the redistribution of electrons when donors are added.

Degenerate Semiconductors (1)

- When the concentration of donor or acceptor atoms increase:
 - excess carrier starts interacting among themselves
 - the discrete donor/acceptor state splits into band of energies

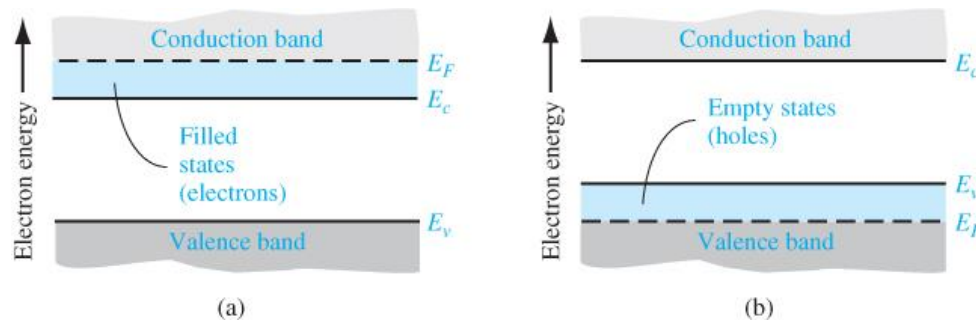


Figure 4.11 | Simplified energy-band diagrams for degenerately doped (a) n-type and (b) p-type semiconductors.

- When the concentration of electrons is more than N_c , Fermi energy is inside conduction band
- When the concentration of holes is more than N_v , Fermi energy is inside valence band
- Such semiconductors are known as degenerate semiconductors
- A degenerate semiconductor shows metallic behavior

Degenerate Semiconductors (2)

- When the band of donor states merge with conduction band, effectively conduction band edge is lowered
- This effect is known as *bandgap narrowing*

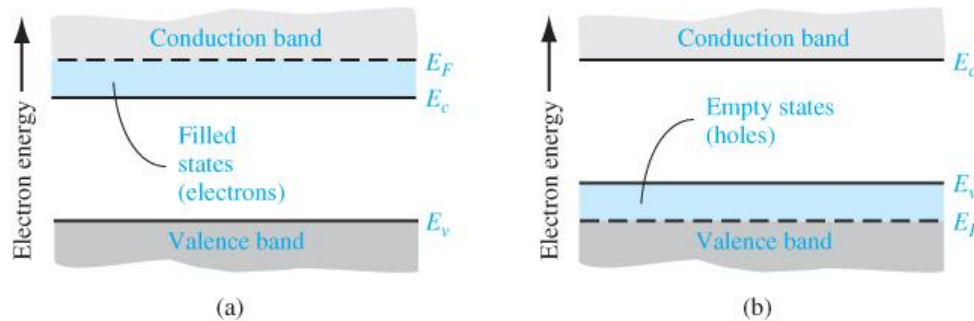


Figure 4.11 | Simplified energy-band diagrams for degenerately doped (a) n-type and (b) p-type semiconductors.

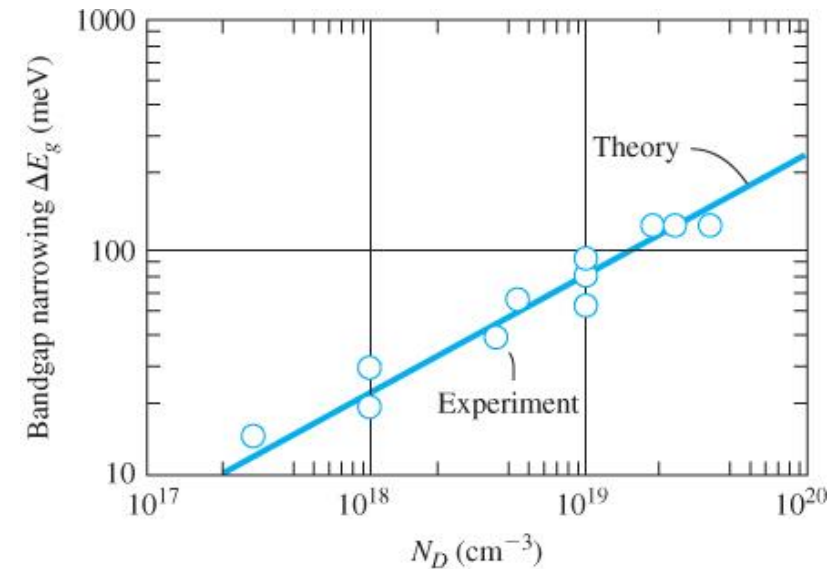


Figure 12.26 | Bandgap narrowing factor versus donor impurity concentration in silicon. (From Sze [19].)

Ionization: Complete and Freeze Out

- At $T = 300\text{ K}$ very few electrons are there in the donor state and almost all go to the conduction band
- Complete ionization occurs

- At $T = 0\text{ K}$ electrons go to the lowest available energy state
- Freeze out occurs

- In the intermediate temperature partial ionization occurs

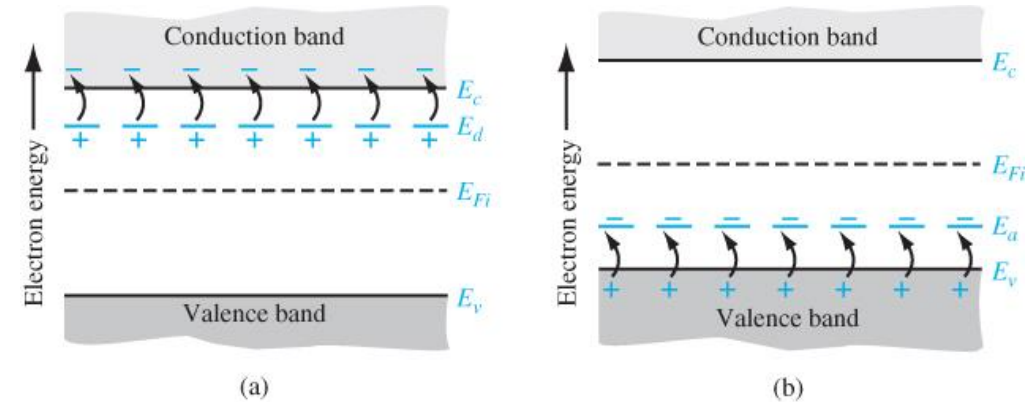


Figure 4.12 | Energy-band diagrams showing complete ionization of (a) donor states and (b) acceptor states.

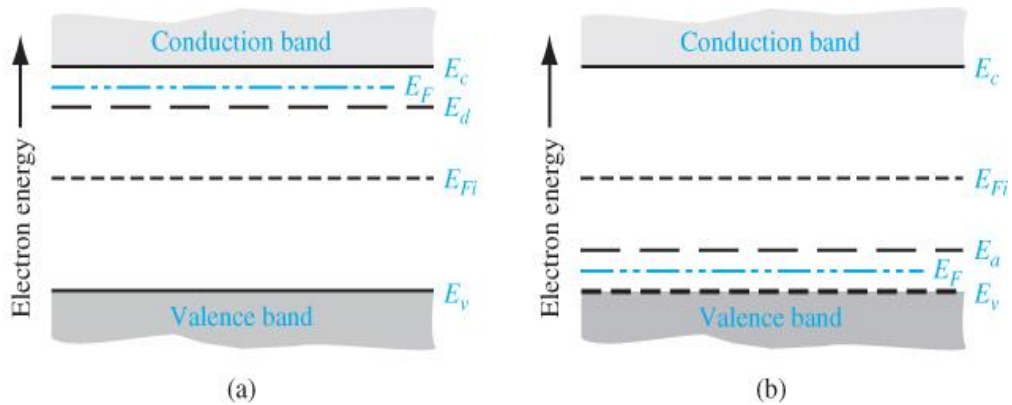


Figure 4.13 | Energy-band diagram at $T = 0\text{ K}$ for (a) n-type and (b) p-type semiconductors.

Electron Concentration vs. Temperature

- Consider an n-type material
- Assume $N_d = 5 \times 10^{14}/\text{cm}^3$
- How will the electron concentration change with temperature

- At $T = 300\text{ K}$, $n_0 = 5 \times 10^{14}/\text{cm}^3$

- Very low temperature partial ionization

- n_i is a strong function of temperature

$$n_i^2 = N_C N_V e^{-E_G/kT}$$

- n_i increases with temperature
- Intrinsic behavior dominate at high temperature

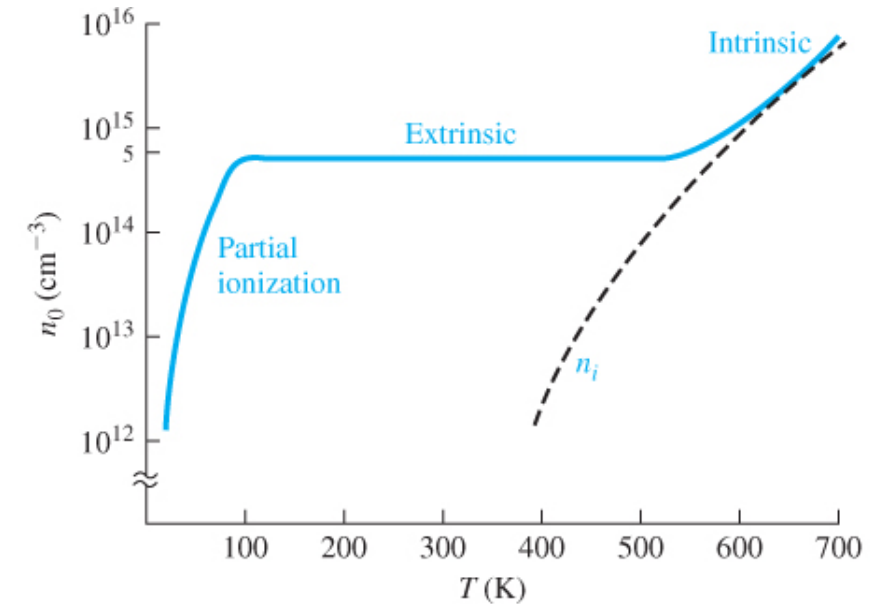


Figure 4.16 | Electron concentration versus temperature showing the three regions: partial ionization, extrinsic, and intrinsic.

Significance of Fermi Energy

- Fermi Energy is a very important parameter for analyzing semiconductor devices

At equilibrium Fermi Energy level is constant throughout the system

- Variation of Fermi Energy with Concentration and Temperature are important

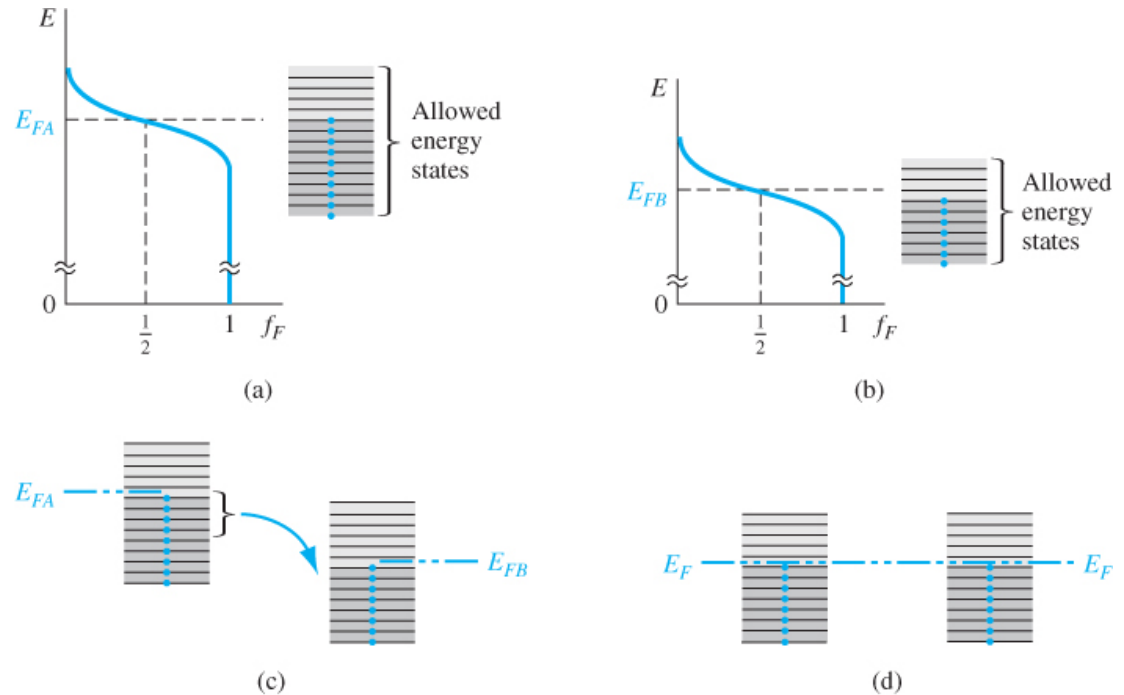


Figure 4.20 | The Fermi energy of (a) material A in thermal equilibrium, (b) material B in thermal equilibrium, (c) materials A and B at the instant they are placed in contact, and (d) materials A and B in contact at thermal equilibrium.

Fermi Energy vs. Doping Concentration

- Electron Conc.: $n_0 = n_i e^{(E_F - E_{Fi})/kT}$
 $\Rightarrow E_F - E_{Fi} = kT \ln \left(\frac{n_0}{n_i} \right)$

- Hole Conc. $p_0 = n_i e^{-(E_{Fi} - E_F)/kT}$
 $\Rightarrow E_{Fi} - E_F = kT \ln \left(\frac{p_0}{n_i} \right)$

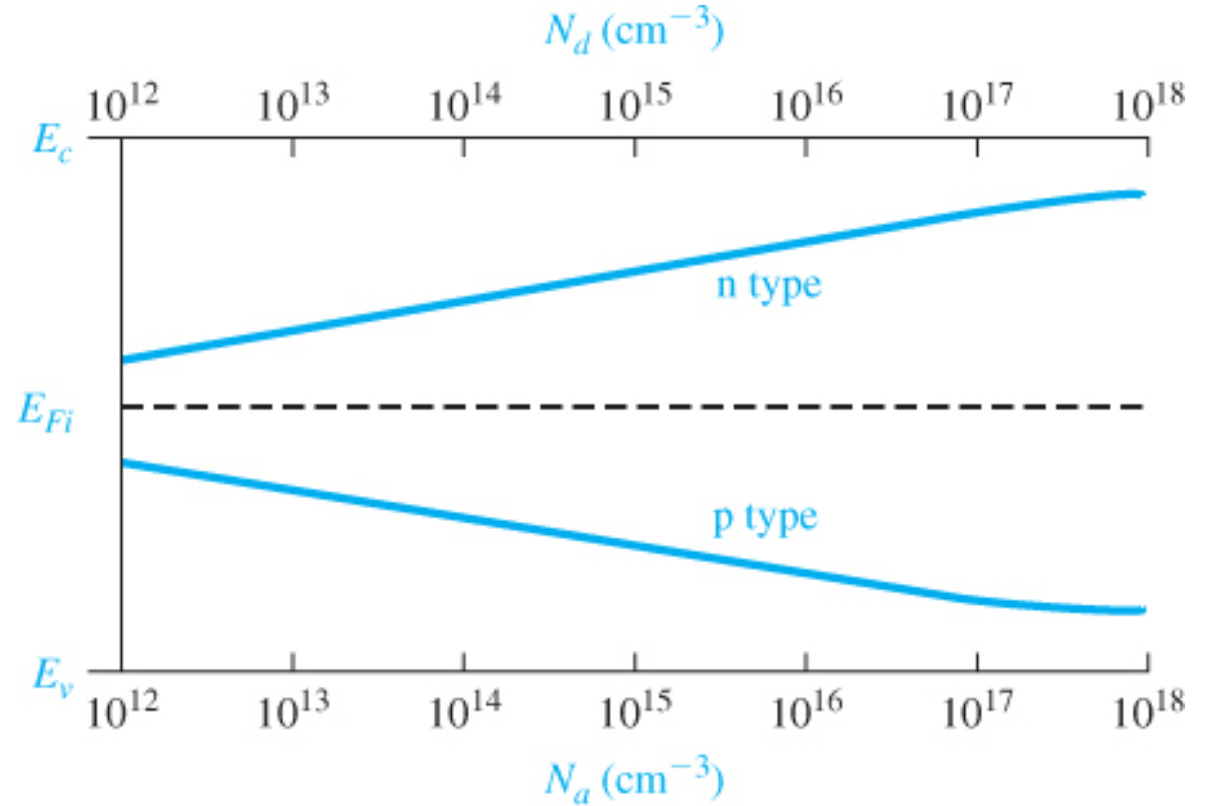


Figure 4.18 | Position of Fermi level as a function of donor concentration (n type) and acceptor concentration (p type).

Fermi Energy vs. Temperature

$$E_F - E_{Fi} = kT \ln \left(\frac{n_0}{n_i} \right)$$

$$E_{Fi} - E_F = kT \ln \left(\frac{p_0}{n_i} \right)$$

- n_i is a strong function of temperature

$$n_i^2 = N_C N_V e^{-E_G/kT}$$

- n_i increases with temperature
- Semiconductor behavior becomes more intrinsic dominated
- Fermi Energy (E_F) comes closer to Intrinsic Fermi Level (E_{Fi})

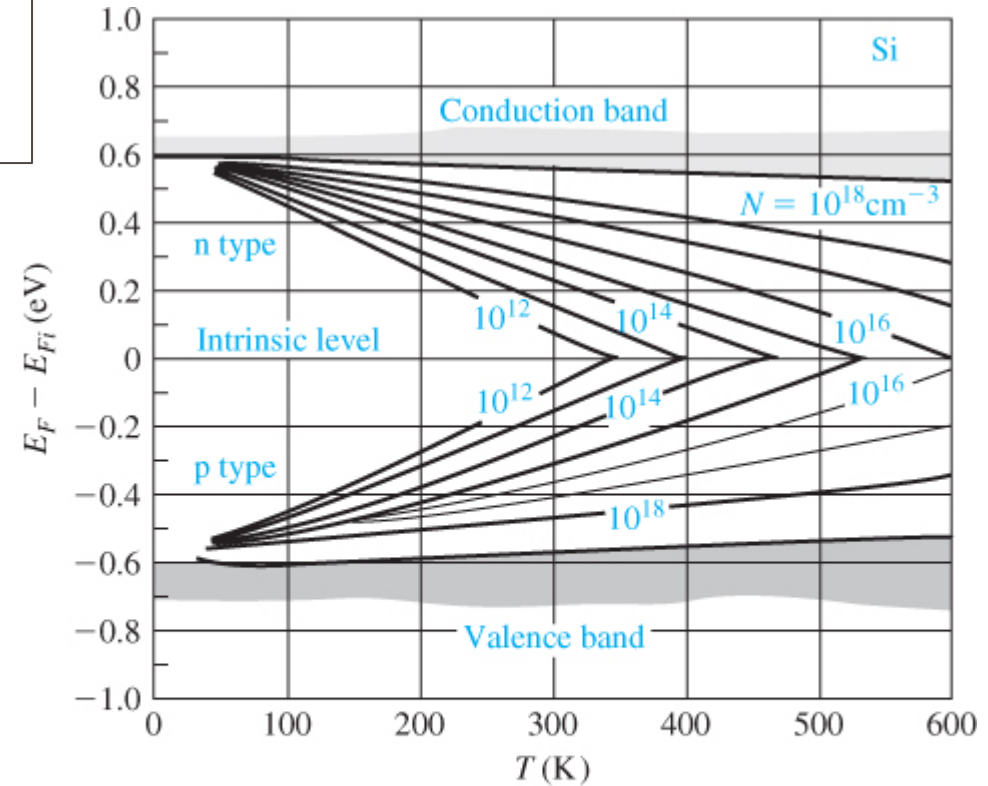


Figure 4.19 | Position of Fermi level as a function of temperature for various doping concentrations. (From Sze [14].)

Carrier Transport

Carrier Transport

- Phenomenon of movement of electrons and holes (charge carriers) is known as carrier transport

Two basic transport mechanisms in semiconductor crystal are:

1. **Drift:** movement of carriers due to electric field
2. **Diffusion:** movement of carriers due to density gradients

Carrier Transport



Drift

Carrier Drift: Mean Time between Collision

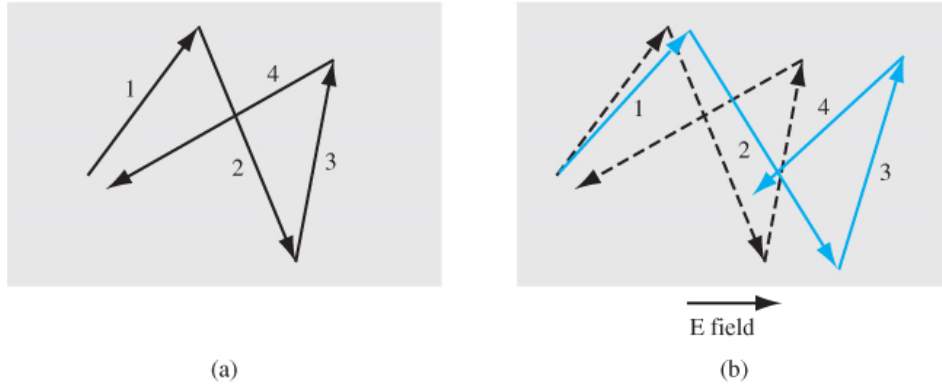


Figure 5.1 | Typical random behavior of a hole in a semiconductor (a) without an electric field and (b) with an electric field.

- A charge carrier is always in **random thermal motion**
 - Of the order of 10^7 cm/s in silicon
- It continues in a path for sometime, hits something, turns around in random direction (scattered),
 - continues on the path hits something, scatters, continues ...
- We can associate mean time between collision with these motion (τ_{cn} and τ_{cp})

- If an electric field is applied, there is a net drift (perturbation) along some direction

Carrier Drift: Mobility

- Acceleration

$$F_{total} = m_n^* a$$

$$\Rightarrow -qE = m_n^* a$$

$$\Rightarrow a = -\frac{qE}{m_n^*}$$

- Average Drift velocity is given as:

$$v_{dn} = -\frac{qE}{m_n^*} \times \tau_{cn}$$

$$\Rightarrow v_{dn} = -\frac{q\tau_{cn}}{m_n^*} \times E$$

- We define mobility of electron μ_n such that $v_{dn} = -\mu_n \times E$ [Valid for *low electric field*]

- Hence, $\mu_n = \frac{q\tau_{cn}}{m_n^*}$

- Similarly for holes:

$$v_{dp} = \mu_p \times E$$

- Hence, $\mu_p = \frac{q\tau_{cp}}{m_p^*}$

Table 5.1 | Typical mobility values at $T = 300$ K and low doping concentrations

	μ_n (cm ² /V-s)	μ_p (cm ² /V-s)
Silicon	1350	480
Gallium arsenide	8500	400
Germanium	3900	1900

Carrier Drift: Scattering Mechanism

Mobility highly depends on the scattering processes

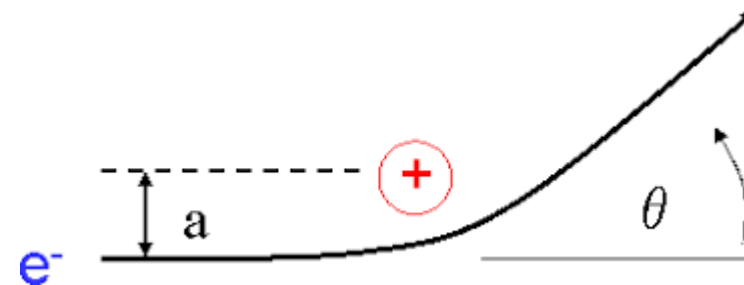
- Scattering process reduces mobility
- Two types of scattering:
 1. Phonon Scattering
 2. Ionized Impurity Scattering

Phonon Scattering:

- Phonons: quantum of vibrational energy that arises from oscillating atoms within a crystal
- Interactions between carriers and phonons (atomic vibrations)

Ionized Impurity Scattering:

- Ionized impurity created by doping and ionization
- Coulomb interaction between carriers and ionized impurities



Mobility: Impact of Temperature and Dopants

- Temperature impacts scattering.
- $\Rightarrow$ Temperature impacts mobility.
- $\Rightarrow$ Temperature impacts most SSD characteristics

Phonon Scattering:

- Phonon scattering increases with increasing temperature
- $\Rightarrow$ Mobility decreases with temperature
- Assuming only phonon scattering occurs:

$$\mu_L \propto \frac{1}{T^{3/2}}$$

Ionized Impurity Scattering:

- Ionized Impurity scattering decreases with increasing **temperature**
 - The random thermal velocity of the carriers increases, and carrier spends less time near ionized impurity
 - $\Rightarrow$ Mobility increases with increasing temperature
- Ionized impurity scattering increases with increasing **impurity concentration**
 - $\Rightarrow$ Mobility decreases with increasing impurity concentration (N_I)
- Assuming only ionized impurity scattering occurs:

$$\mu_I \propto \frac{T^{3/2}}{N_I}$$

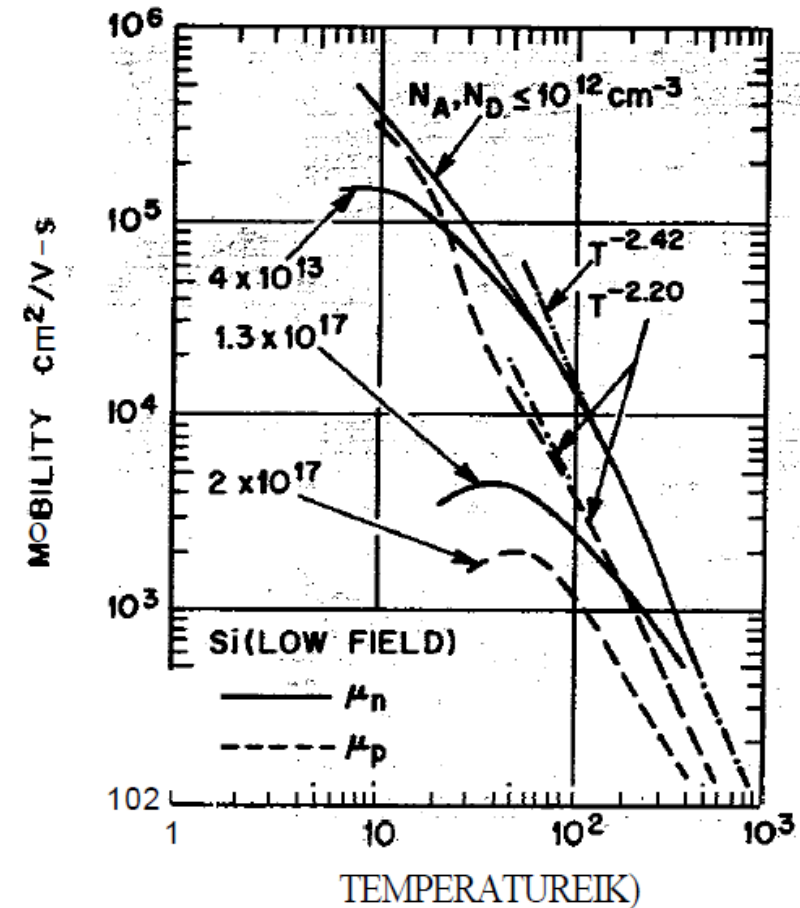
Mobility: Combined Effect

- Assuming only phonon scattering occurs:

$$\mu_L \propto \frac{1}{T^{3/2}}$$

- Assuming only ionized impurity scattering occurs:

$$\mu_I \propto \frac{T^{3/2}}{N_I}$$



Carrier Drift: Current Density (Electrons)

Problem: Consider a semiconductor bar of dimension $A \times L$ shown alongside. Assume that the electron density is n and average drift velocity is v_{dn} .

- Consider an electron that enters the bar and leaves it with average drift velocity of v_{dn} . Compute the current due to this electron. Charge on an electron is $-q$.
- Compute the total current for the bar due to electron density n .
- Compute the current density for the bar.

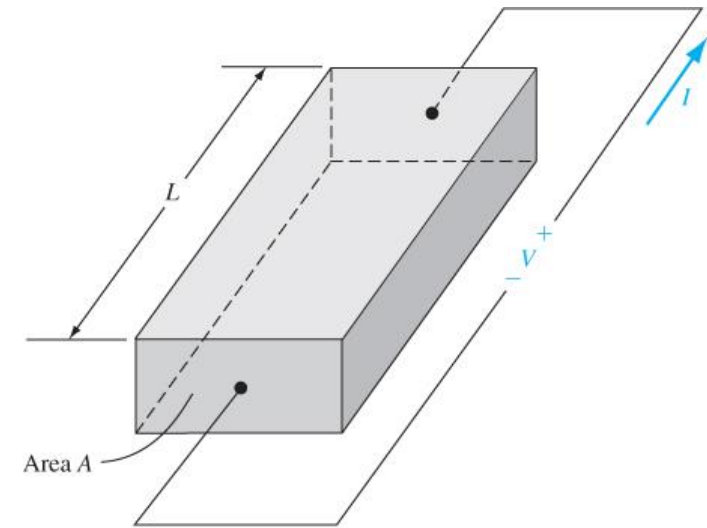


Figure 5.5 | Bar of semiconductor material as a resistor.

a) Time take $t = \frac{L}{v_{dn}}$

$$\text{Current} = \frac{\text{Charge}}{\text{Time}} = \frac{-q \times v_{dn}}{L}$$

b) No. of electrons $N = A \times L \times n$

$$\text{Total Current} = \frac{-q \times v_{dn}}{L} \times A \times L \times n$$

c) Current density

$$= \frac{\text{Current}}{\text{Area}}$$
$$= -q \times v_{dn} \times n$$

Carrier Drift: Current Density (Total)

- Current density due to electrons
 $= -q \times v_{dn} \times n$

- For holes we can derive similarly
- Current density due to holes $= q \times v_{dp} \times p$

For electrons: $v_{dn} = -\mu_n E$

For holes: $v_{dp} = \mu_p E$

- Current density due to electrons
 $= q\mu_n En$

- Current density due to holes
 $= q\mu_p Ep$

When both electrons and holes are present

- Total current density due to drift $= J_{drift} = q(\mu_n n + \mu_p p)E$

Carrier Drift: Problem

Consider a GaAs sample with $N_a = 0$ and $N_d = 1 \times 10^{16} \text{ cm}^{-3}$.

Calculate the drift current density when the applied electric field is $E = 10 \text{ V/cm}$.

Assume complete ionization.

$$n_i = 1.8 \times 10^6 \text{ cm}^{-3}$$

$$\begin{aligned} n &\approx 1 \times 10^{16} \text{ cm}^{-3} \\ p &= 3.24 \times 10^{-4} \text{ cm}^{-3} \end{aligned}$$

Table 5.1 | Typical mobility values at $T = 300 \text{ K}$ and low doping concentrations

	$\mu_n \text{ (cm}^2\text{/V-s)}$	$\mu_p \text{ (cm}^2\text{/V-s)}$
Silicon	1350	480
Gallium arsenide	8500	400
Germanium	3900	1900

$$\begin{aligned} J_{drift} &= q(\mu_n n + \mu_p p)E \approx q\mu_n nE \\ &= 1.6 \times 10^{-19} \times 8500 \times 10^{16} \times 10 \\ &= 136 \text{ A/cm}^2 \end{aligned}$$

Conductivity and Resistivity

- Total current density = $J_{drift} = q(\mu_n n + \mu_p p)E$
- We can write: $J_{drift} = \sigma \times E$ where
$$\sigma = q(\mu_n n + \mu_p p)$$
- We call σ as *conductivity*
- It is a function of mobility and carrier concentration

- Resistivity is defined as the reciprocal of conductivity

$$\rho = \frac{1}{\sigma}$$

- Ohms law can be derived using $E = \frac{V}{L}$

Velocity Saturation

For electrons: $v_{dn} = -\mu_n E$

For holes: $v_{dp} = \mu_p E$

- Above equations suggest, velocity goes on increasing with electric field
 - Practically, velocity saturates
 - What happens if more electric field is applied after velocity saturation?
 - Carriers lose energy through increased levels of interaction with the lattice (by emitting phonons)
-
- For silicon, velocity saturates to $1 \times 10^7 \text{ cm/s}$ at 30 kV/cm
 - Current density also saturates (does not increase with electric field)

References

- D. A. Neaman, *"Semiconductor Physics and Devices"*